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Low Light Enhancer: A Low Light Image Enhancement Model Based on U-Net using Smartphone

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***Abstract* - In this paper, we propose an improved way low light image-enhancing model for Smartphone devices by training a U-Net based fully convolutional neural network (CNN) with raw sensor images. We collected a new dataset captured via a Smartphone camera which is a set of low light images, corresponding long exposure images are captured for ground truth. Our primary goal is to train our model with different Smartphone camera devices having a Bayer filter with different image sensors. The second goal is to get noise-free images with minimum color distortion.**

Keywords: Low Light Image Enhancement; Fully Convolutional Neural Network; Raw sensor images; Bayer sensor.

I. INTRODUCTION

Computer vision and digital image processing technology are rapidly evolving and being used in almost all sectors of our daily life like video monitoring, intelligent transportation, remote sensing monitoring, industrial production, and military application. To get the proper information from an image, the image needs to be properly illuminated because of different conditions like nighttime, cloudy days, indoor, lack of external lights. Again for mobile camera devices, the image is less qualitative than a DSLR due to limitations like small sensor size, compact lenses, and lack of specific hardware.

Low light refers to the environment that has low illumination and does not match the normal environmental situation. Due to the capture of low luminance images that images suffer from brightness, contrast, gray range, color distortion, and as well as substantial noise.

To improve the result of captured images in low-light environments, researchers have sought many ways from both hardware and software. There are many low-light image enhancement methods. Such as, CNNs models are being used as the most basic deep learning frameworks in many research works [1]-[6]. Chen *et al* [1] proposed a dataset of raw short exposure low light images i.e., See-In-The-Dark Dataset (SID) with corresponding to long exposure images and developed a pipeline to process these images based on a Fully Convolutional Neural network (FCNN). The results were more improved than the traditional methods. However, they had some limitations like large dataset, training accuracy, no human images in the dataset, and did not generate High Dynamic Range (HDR) images.

Again based on Retinex theory, Shen *et al.* [2] proposed a method using an MSR network (MSR-net) which was also based on a CNN architecture, while Guo *et al.* [3] introduced a pipeline consisting of a de-noising net and low light image enhancement net (LLIE-net). Wei *et al.* [4] collected a Low-Light (LOL) dataset containing low light images corresponding to their normal light images and proposed a deep network named RetinexNet. Later on, Li *et al.* [5] introduced a network called LightenNet that takes the low-light image as input and outputs an illumination map which is used to generate an enhanced image. Afterward, Zhang *et al.* [6] constructed an effective network i.e Kindling the Darkness (KinD) network, which consists of a layer decomposition net, a reflectance restoration net and an illumination adjustment net to train a pair of images of different exposures. Liba et al. [7] developed a system using a novel combination of motion-adaptive burst capture, learning-based white balance, robust temporal de-noising, and tone mapping to create high-quality photographs in low light on a handheld mobile device. Lore et al. [8] proposed a deep auto-encoder-based method for identifying signal features from images captured in a dark environment and adaptively enhancing the brightness of images without over-amplifying the darker parts in images with a high dynamic range. This results in significant reliability of the approach with various image enhancement techniques. Syed Waqas Zamir et al. [7] proposed a data-driven approach based on a novel loss function that is capable of generating well-exposed sRGB images with the desired attributes: sharpness, color vividness, good contrast, noise-free, and no color artifacts. However, their work contained the imperfect ground truth of the dataset, and therefore the learned network was partially optimal. Chen et al. [10] showed that photographic images can be synthesized from semantic layouts by a single feed-forward network with appropriate structure, trained end to end with a direct regression objective. But their work was not property ready to achieve perfect photorealism.

By analyzing the present state of the arts, it is confirmed that the deep learning methods showed outstanding performance in low light image enhancement. However, the major drawbacks are the large datasets, time complexity, and not enough publically available data designed specifically for low-light image processing. After considering all the previous works and their limitations we tried to contribute in this field by introducing our CNN model and building a dataset with a set of low light images taken under an extremely low light environment via a Smartphone camera device. We trained our model by a set of raw short exposure images with the ground truth image as a raw long exposure image. Low-light image enhancement is effective with raw sensor data and most of the devices use Bayer filter in their sensor. So we collected the raw images and packed the images in according to their Bayer filter pattern. Then we fed them to the network.

After training, we used the image pair and the output image of the network for post-processing which is done by High Dynamic Range (HDR) tone mapping which is a technique that produces a dynamic range of luminosity in a standard image. We did it by tone mapping the images. Four images were used-the input image, the output image, the scaled image, and the long exposure image. Combining them the tone is mapped across the output image pixels, which saves our image from being overexposed or underexposed.

The body of this work is divided into the following sections. The associated proposed model is described in Section II. The experimental result of the proposed model is presented in Section III. Section IV brings the paper to a conclusion.

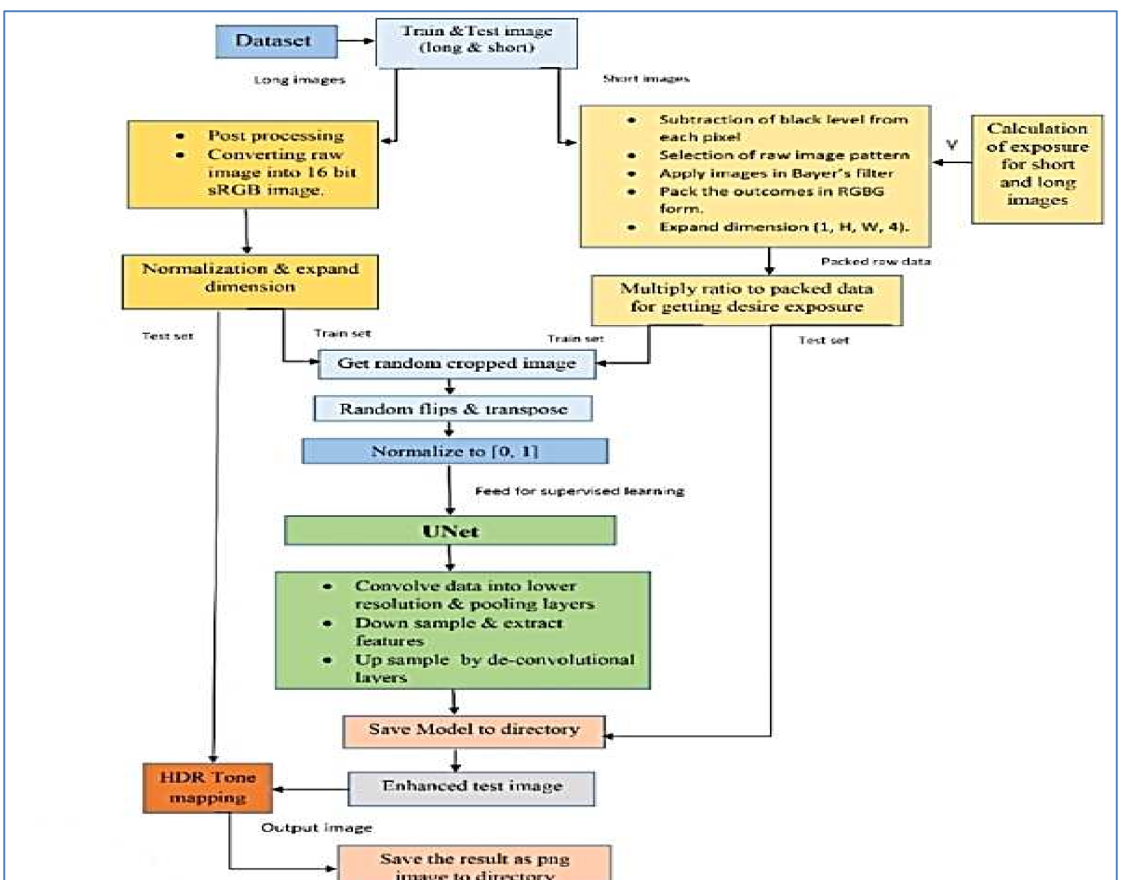

Fig. 1. The workflow diagram of the proposed model.

II. PROPOSED MODEL

Through our Night Light Enhancement process, we fed short exposure image(s) with corresponding long exposure images as ground truth from the Night Light Enhancer dataset. First, we pack the raw data from the imaging sensor to perform preprocessing and get randomly cropped outcomes with the help of normalized corresponding reference images. For supervised learning, we feed the input to U-net and perform post-processing to the outcome to get our desired HDR image.

Our dataset contains short and long raw sensor data in dng format. Multiple short exposure images can correspond to only one long exposure image. The proposed model of our paperwork can be described in three different phases i.e., preprocessing phase, network, and post-processing phase. The workflow diagram is presented in Fig. 1.

The raw sensor data is present as a Bayer array that is first pre-processed into separate input channels for each pixel color and is spatially reduced in both dimensions. The purpose of down-sampling the Bayer array through pre-processing is to sort the input data in a more optimized way for processing, but there is no data loss in this step. This is because the first layers of the U-Net will eventually work on this data and performs better with four smaller channels with separated colors rather than one larger image channel. The next task in preprocessing is to remove the black levels from the optimized data. An amplification factor is multiplied to specify the desired level of brightness but it is applied only after subtracting the black level unless the black level will also amplify and produce an unwanted black effect in the image. The proposed model for light enhancement is presented in Fig. 2.

For training, the preprocessed data is then fed into the U-Net. We used U-Net which is encoder-decoder architecture for skip connections. Here, U-Net takes in raw camera sensor data from dark images and learns to predict what the image would look like with better lighting. The technique works by learning from the raw camera data in a low-light image and predicting each pixel of the image taken from the same angle with more lighting. The U-Net contains a series of layers that's are repeatedly convolved the data into lower resolutions using pairs of convolution and max-pooling layers while extracting progressively more detailed feature information. The convolutional layer component down-samples the data and sends it to a subsequent Convolution Layer for further down-sampling, while also extracting the features. The convolutional layers use the "same" padding, so, they do not change the resolution of the feature maps, unlike the de-convolution layers.

The utmost down-sampled data is sent to a De-convolution Layer component for up-sampling. The up-sampled image, along with the feature from the corresponding resolution of down-sampled data is concatenated, and the output is fed into a subsequent convolution layer. The last layer of the U-Net makes use of a sub-pixel up-sampling layer to ultimately produce a brightened image from the original raw sensor data. The U-Net uses the corresponding sRGB images during training, from which ground truth has to determine and perform the supervised learning.

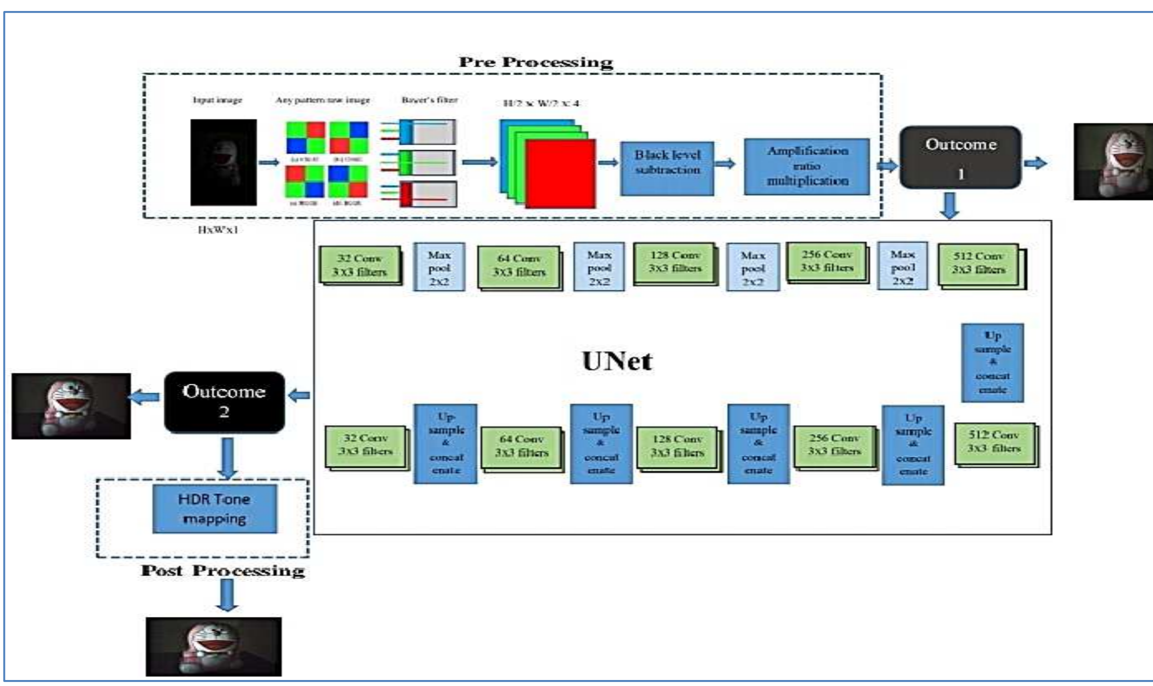

Fig. 2. Proposed model for low light enhancement.

III. EXPERIMENT RESULTS

A. Dataset

The dataset we used in our approach is a collection of mobile phone captured RAW images. In the dataset, there is a burst of low ISO images with corresponding high ISO images. The data was preprocessed by subtracting black levels and packing the raw sensor data into four channels (RGBG) from their individual Bayer patterns (RGBG, RGGB, BGGR, or GRBG). RawPy library makes this process easier.

Our dataset contains 1030 raw short exposure images with a reference of a long exposure image. We used the mobile phone camera to capture our images in extremely low light in both indoors and outdoors. The exposure range for long is 1/3 for normal light images and 1/400 to 1/500 for the outdoor images. Again Exposure for short is 1/60 to 1/100 for normal light image and 1/400 and 1/10000 for daylight image. ISO range is from 100 to 200 for long exposure images and 1600 for short exposure images. Our dataset contains human images. Again a dataset of low light captures on mobile phones is not publicly available on the internet. 80% of our data is for training and 20% for testing which are selected randomly. Some short and long exposure images are presented in Fig. 3.

Fig. 3. The low light short exposure images are in the back and the corresponding long exposure images are shown in front.

B. Loss Function and Optimizer

In this work, the default architecture of U-Net is used for training our model. For optimization of the models, an empirical learning rate is used i.e., 0.0001. As the Adam optimizer is an adaptive optimizer it takes less time to train the model, in this work, we consider the Adam optimizer to train the model. The test data is used to see how well the model responds. The L1 loss function is used to minimize the error and is the total of all the absolute deviations between true and expected values. The loss function equation is presented in (1).

$$L(y, \hat{y}) = \sum_{i=1}^{n} | y - \hat{y}_i | \quad (1)$$

Here y is the target value and $\hat{y}$ is the estimated value from the trained model. It is minimizing the sum of the absolute differences between the target value (y) and the estimated values ($\hat{y}$). We trained our model using L1 loss and Adam optimizer. The learning rate is initially set to 0.0001. It is reduced to 0.00001 after training the model for 2000 epochs. We continued the training process for 4000 epochs. The loss curve of the model is shown in Fig. 4.

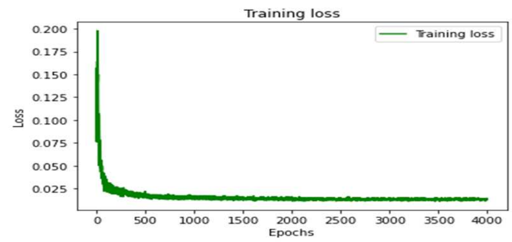

Fig. 4. Loss curve of the proposed model.

C. Experiment Result on Indoor Images

In this section, we experimented with how our model performs on indoor images from three different datasets where sample 1 is from the Night Light Enhancer dataset, sample 2 is from the SID dataset, and the image in sample 3 is captured with an iPhone camera. These indoor images were taken without any external lights. In Fig. 5, 5(a) is the input image taken in a dark environment, 5(b) is the output of our trained model and 5(c) is the image after post-processing of the output image.

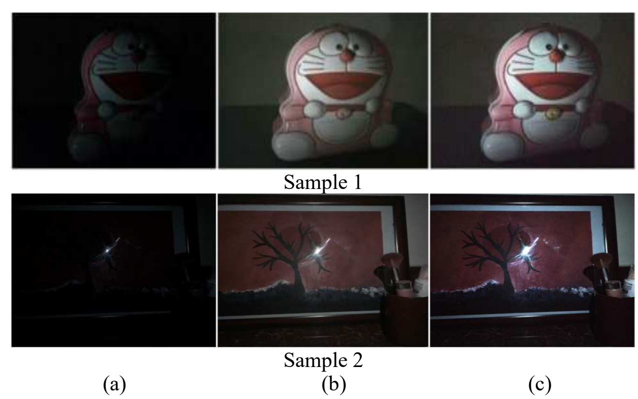

Fig. 5. Experimental result images: a) Input image, b) Output image after training the model, and c) Post-processed image of the output image.

The input image Fig. 5(a) is in .dng format and the raw patterns of these images are BGGR. The image Fig. 5(b) is the result we achieved after training our model. In Fig. 5(b) the noise is reduced and the background is properly visible. Again the colors of the images are properly restored but the brightness is not mapped in equal proportion. Sample images in Fig. 5(c) are post-processed images. We took the exposure from Fig. 5(a) and Fig. 5(b) and applied it in Fig. 5(c) for proper brightness and color tone mapping.

D. Experiment Result on Outdoor Images

In this section, we experimented with how our model performs on outdoor images. These outdoor images were taken without any external lights. In Fig. 6, Sample 1 is the image from Oppo camera devices and Night Light Enhancer Dataset. These images were taken in night light without any eternal light sources which are in .dng format. Fig. 6(a) is the input image taken in a dark environment, Fig. 6(b) is the output of our trained model, and Fig. 6(c) is the image after post-processing the output image. The input image in Fig. 6(a) is in raw format and we retrieved the raw pattern from these images and packed them into RGBG format. The image in Fig. 6(b) is the result we achieved after training our model. In Fig. 6(b) the noise is reduced and the background is properly visible. Again the color of the images is properly restored but the brightness is not mapped in equal proportion. Images in Fig. 6(c) are a post-processed image. We took the exposure from the image of Fig. 6(a) and Fig. 6(b) and applied it in Fig. 6(c) for proper brightness and color tone mapping.

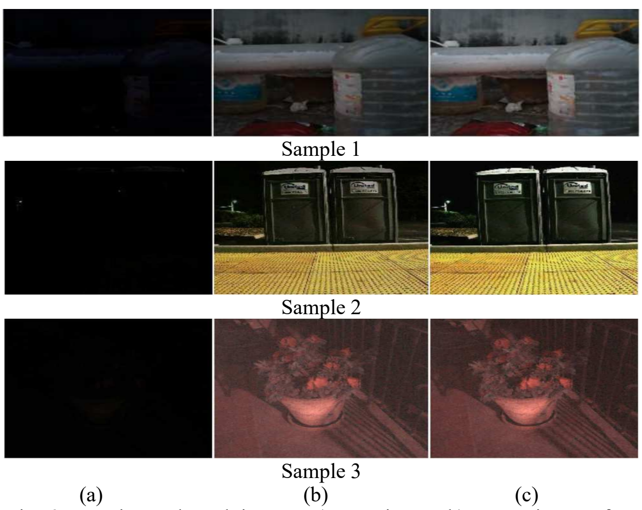

Fig. 6. Experimental result images: a) Input image, b) Output image after training the model, and c) post-processed image of the output image.

E. Post-processing: HDR Tone mapping

In Fig. 7, we showed the post-processing algorithms applied for HDR tone mapping. Firstly we took the exposure time from our input and output image (from our trained model). Then we merged the exposure time sequence into one HDR image. In OpenCV there are two methods i.e., a) Debevec algorithm and b) Robertson algorithm. Both are applied in an image, and the result is shown in Fig. 7(a) and Fig. 7(b) respectively. As it includes the maximum dynamic spectrum of all exposure images, the HDR image is of type float32 and not uint8. We mapped the 32-bit float HDR data into the range[0..1] in all of these methods, as in some instances the values can be greater than 1 or less than 0, so we clipped the HDR data to prevent overflow. However, the results we get both from Fig. 7(a) and Fig. 7(b) is not expected. So we applied another method i.e., Martens Fusion algorithm, where we do not need the exposure time from either input or output image rather we merged the exposures from images. Here, we did not apply any tone map algorithm because marten's algorithm already gives the output as [0..1] range. The result of our approach is shown in Fig. 7(c).

(a) (b) (c)

Fig. 7. Post-processing result images after applying HDR tone mapping algorithms: a) Debevec algorithm, b) Robertson algorithm, and c) Martens Fusion algorithm as our approach.

F. Comparison Proposed Model with Present State of the Art

In this section, we compare the proposed model with the present state-of-the-art image processing techniques. For that, our model is compared with two other methods, which are, DM3D de-noising and contrast stretching methods. The state-of-the-art methods provide results that differ for specific cameras and different environments. Sometimes, these methods provide much complexity with respect to the others. However, these limitations are covered by the proposed model which is an end-to-end model for direct image processing for low light images.

We applied the traditional BM3D technique in Fig. 8, sample 1. As we see, the output image of sample 1(b) in Fig. 8 is visible but the amount of noise is high which is measured by the value of Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). The respective PSNR and SSIM values for Sample1, Sample2, and Sample3 are presented in Table 1. In sample 2 we applied another image processing technique i.e., contrast stretching. As result in sample 2(b) the output image is not properly restored and the image has color distortions but the noise is less. In sample 3 we showed the result of the output from the trained model. Our model output is better than the state of the art. Again the color is also properly restored.

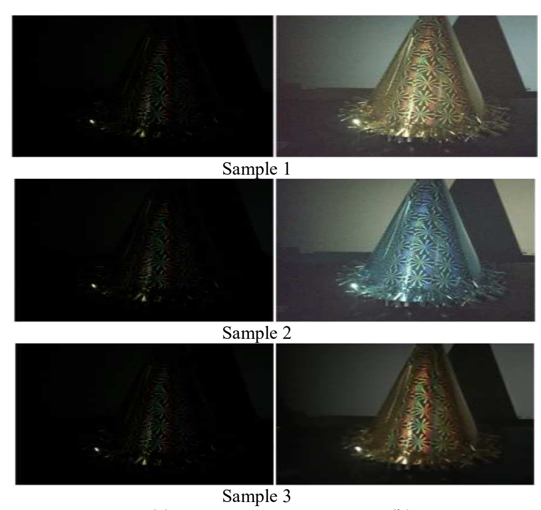

Fig. 8. Comparison with the present state of the art: a) Input image, and b) output image.

TABLE 1: COMPARISON OF IMAGE PROCESSING TECHNIQUES FOR DIFFERENT STATE-OF-THE-ART METHODS

Sample	Methods	PSNR and SSIM Value
Sample 1	BM3D de-noising	PSNR = 27.7098766 SSIM = 0.034
Sample 2	Contrast Stretching	PSNR = 27.7075516 SSIM = 0.035
Sample 3	**Proposed model**	PSNR = 27.8981825 SSIM = 0.282

G. Controlled Experiments

The accuracy of the given model in terms of PSNR and SSIM Ratio is shown in Table 2. Here, the first row shows the Structural Similarity. The second row records the U-Net outcome. We compared the proposed model's result with various state-of-the-art models' architectures. As U-Net shows higher PSNR, we replaced CNN with U-net, however, CNN has higher SSIM, it often suffers from color loss. After the study, we discovered that working directly on raw sensor data reveals much more performance instead of sRGB images in intense low light conditions. The PSNR and SSIM of raw sensor images are seen in the third row of Table 2. We used the L1 loss function; however, we apply the other loss functions instead of L1 to measure the effectiveness of L1 loss. From, the results are shown in Table 2, it is revealed that L1 gives better accuracy than L2 loss. Adding a total variation loss does not improve accuracy. Table 2 row 4 shows the L1 loss PSNR and SSIM values. From the experiment, it is revealed that the RGB images that are output from the model are not properly mapped. For that, here, we introduce the HDR tone mapping to adjust the overall brightness of the image. The PSNR and SSIM values of HDR image are shown in Table 2 row 5.

TABLE II: CONTROLLED EXPERIMENTS REPORTS MEAN PSNR AND SSIM IN EACH CONDITION

Condition	PNSR	SSIM
Our default pipeline	28.88	0.787
U-net	27.40	0.79
Raw	17.40	0.554
L1 loss	28.47	0.784
HDR	29.318	0.363

H. Discussion

The proposed model produces good results for small smart camera devices which are usually more prone to noise. It does not lose any information and give good PSNR and SSIM scores. The proposed method is performed more effectively by using the HDR tone mapped as post-processing at the output image of the network. If we try to make the network learn the processing of a long exposure image, the performance of the network falls significantly as it has to learn histogram. For that reason, we processed the raw image into sRGB.

IV. CONCLUSIONS

Based on the limitations of SID methods, we try to improve their model with a new dataset for low light image enhancement to ensure appropriate balance among several factors such as image color, visual effect, and information entropy. The proposed model produces good results for small smart camera devices which are usually more prone to noise. It does not lose any information and give extremely good PSNR and SSIM scores. Here, the resulting HDR output is much mapped. The experiments are compared between U-Net and CNN, where U-Net shows much better results. There is a huge scope for future works such to work with video as input and build a dataset that are capture any dynamic objects.

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